

Vizualink: Evaluating Usability and Loyalty Impacts of Text-Graph Integration in Data-Driven Articles

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ABSTRACT

Data-driven articles have emerged as a powerful method for transforming complex datasets into accessible narratives. Existing approaches, such as interactive visualizations and scrollytelling, enhance engagement and simplify complexity by guiding users through data stories. However, these designs often prioritize narrative structure over reader-driven exploration, which may limit opportunities for autonomous interaction. To address this limitation, existing literature explored features that dynamically link textual content to corresponding elements in interactive graphs. In this study, we investigate the impact of such tools on the reading experience, specifically the hedonic and pragmatic dimensions, in addition to loyalty intentions. We implemented Vizualink, a text-graph linking feature, and examined its impact through an online study with 360 participants. Our results indicate that Vizualink enhances the reading experience, which in turn positively impacts readers' intention to return for more content, recommend the article, and consider paying for access. A thematic analysis of participant comments highlighted enhanced interactivity and criticism regarding potential hindrances to user experience.

Index Terms: Vizualink, Text-Graph Integration, Data-Driven Articles, Interactive Systems

1 INTRODUCTION

Data-driven articles have become a vital method for transforming intricate datasets into accessible stories, utilizing digital platforms to present increasingly interactive content [38]. These articles typically combine narrative text with embedded quantitative data and visualizations, such as charts, graphs, or maps. By grounding articles in concrete data, they enhance credibility and offer readers a transparent basis for the claims presented [32, 27]. Despite their strengths, data-driven articles face a key challenge: they often contain dense content filled with complex statistics and analyses that are difficult for readers to understand [24, 40].

Over time, various strategies have been adopted to tackle this complexity. Infographics and interactive visualizations have become widely used tools to simplify data and enhance storytelling by turning raw information into compelling narratives [41]. These techniques improve reader engagement, making data both more accessible and easier to interpret [26, 7]. Another innovative approach is scrollytelling—a method that integrates storytelling with interactive visuals that evolve as users scroll, providing a more immersive and guided experience through complex narratives [36]. The rise of tools like Flourish [15] and Datawrapper [12] has further advanced these methods, enabling journalists to create and distribute interactive visualizations and data-rich stories more efficiently [9].

Additionally, prior work has also explored the creation of text-graph links that dynamically link textual content to corresponding

elements in interactive graphs [38, 31], allowing more exploration rather than passive consumption [37]. The existing literature contains user studies evaluating the impact of such tools on aspects like guiding reading [38], precision and recall [31], annotation accuracy [22, 39], and understanding [3].

However, less attention has been given to other important factors influencing reader satisfaction, such as hedonic and pragmatic aspects of the reading experience and loyalty intentions. The hedonic aspect refers to how enjoyable and stimulating the reading experience is, while the pragmatic aspect concerns the ease and efficiency with which readers can interact with the content [19]. Loyalty intentions capture readers' willingness to return to the site, recommend it to others, or pay for access [30]. These aspects are particularly important as previous literature consistently highlighted them as key drivers of reader engagement and understanding [17, 23].

In this paper, we address this gap by implementing a text-graph linking feature, Vizualink, and studying its impact on these dimensions. We evaluate Vizualink through an online study with 360 participants. This context is well-suited due to the widespread use of data-driven articles, especially since the COVID-19 pandemic, and the importance of audience loyalty in a news industry that increasingly relies on audience-supported revenue models [27]. In this paper, we aim to address the following research question:

RQ: How does the integration of interactive text-graph links in data-driven articles impact the reading experience and loyalty intentions?

To address our research question, we adopted a mixed-methods approach that combined qualitative and quantitative data collection and analysis. Specifically, we applied a design thinking approach [4] to explore the integration of Vizualink in data-centered articles. This approach emphasized iterative prototyping and user-centered feedback to refine text-graph integration. The contribution of this paper consists of implementing an interactive text-graph linking feature within a data-driven article published by an online newspaper, and empirically validating a model assessing its impact on reader experience and loyalty.

2 BACKGROUND & RELATED WORK

Several research efforts have focused on linking textual narratives and visual data representations to enhance the creation and reading experiences of data-driven articles. For instance, a no-code authoring tool, VizFlow, which leverages text-chart links to help authors create dynamic data-driven articles in formats like scrollytelling, was developed by Sultanum et al. [38]. They showed that VizFlow enabled rapid and expressive article creation. Readers preferred reflowed articles with text-chart linking over static ones, while authors appreciated the ease of use and flexibility offered by the system. Similarly, a method to automatically link phrases in text to elements in bar, line, and scatter plots, called ChartText, was proposed by Pinheiro and Poco [31]. Evaluations on web and academic datasets showed ChartText achieved moderate to high accuracy in the creation of these links, demonstrating the potential for improving reading experiences. In a related direction, an automatic annotating technique that links textual descriptions to specific areas

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in visualization images using object detection and natural language processing was also presented by Lai et al. [22]. Their tool, Vis-Annotator, generates step-by-step animations that guide the audience's attention by highlighting described elements. A user study showed that the method improved the efficiency and clarity of data presentations, though some challenges remained in NLP accuracy. Expanding the focus to authoring support, Sultanum et al. [39] introduced DATATALES, a prototype system that uses large language models to assist in authoring data-driven articles by generating linked narratives based on charts and user annotations. Their findings showed that the system helped users generate initial drafts, supported error checking via text-chart linking, and inspired new story directions. These studies developed innovative tools to enhance the authoring and reading of data-driven content, evaluating their impact on aspects such as reading guidance, information retrieval accuracy, annotation precision, and understanding.

However, usability factors such as the hedonic and pragmatic dimensions of the reading experience, as well as loyalty intentions, remain to be further explored. In contexts such as online newspapers, where switching providers entails minimal costs [33], these metrics play a substantial role in customer retention and overall business success [17]. Prior work has shown that a better reading experience leads to higher loyalty by enhancing user interactions, particularly through elements like interactivity [2]. Specifically, the pragmatic dimension of the reading experience plays a crucial role in influencing reader loyalty by facilitating task completion, leading to increased willingness to return and purchase intentions [21, 20]. Additionally, the hedonic aspect of the reading experience plays a significant role in increasing user loyalty towards news websites [5]. When users find a news website enjoyable and engaging, they are more likely to be satisfied with their experience, which in turn strengthens their loyalty and intention to revisit the website [14]. Based on these findings, we formulate the following hypotheses:

H1: Text-graph integration in a data-driven article enhances the usability dimensions of the reading experience.

H2: A more positive reading experience is associated with higher loyalty intentions.

3 SOLUTION DESIGN

We designed Vizualink to create a link between textual information and graph components, with the aim of assessing the impact of such features on the reading experience and loyalty intentions in online data-driven articles. Vizualink creates a connection between the narrative and the data visualization by allowing readers to hover over a highlighted textual element to see the corresponding part of the graph (see Figure 1). Textual data points are highlighted using colored spans to emphasize key information that is presented in the text and refers to a graph. When a user hovers over one of the highlighted data points, Vizualink emphasizes the corresponding part on the graph. The prototype of Vizualink was developed using JavaScript, which we used to extend the functionalities of the data visualization framework Datawrapper, enabling dynamic highlight features and visual emphasis without altering the original graph structure. The solution dynamically modifies visual and textual elements within articles.

4 EXPERIMENTAL SETUP

We carried out an online experiment to evaluate the impact of Vizualink on the reading experience. The article used for this assessment was drawn from a collection of over 600 data-driven articles compiled for a separate study. We chose an article published in a Swiss online newspaper during the COVID-19 pandemic due to its rich visual and numerical content, including six graphs and hundreds of data points. This made it a good test case for examining Vizualink's usability and its potential influence on loyalty in-

tentions, such as revisiting the newspaper for similar content. The original article was written in French, and the entire experiment, including all materials and interactions, was conducted in French.

We recruited participants through the Prolific platform, and they completed the survey via Qualtrics. The university's ethics board approved the experiment. The experiment included 360 participants who are fluent in French, each earning approximately 12 USD per hour for their participation. On average, participants took 12 minutes to complete the experiment. The participant pool was composed of 48% females, 51% males, and 1% non-binary individuals. Participants were randomly assigned to one of three groups at the start of the survey. Group 1 (hereby referred to as *the Control group*) accessed the original online article without any enhancements. Group 2 (hereafter referred to as the *Highlights group*) viewed the article with all textual data points related to the graphs highlighted; however, hovering over these highlights did not trigger any interaction with the graphs. Group 3 (hereby referred to as the *Vizualink group*) experienced the full functionality of Vizualink, including highlighted textual data points and the ability to hover over them to reveal corresponding sections in the graphs. This incremental design aimed to isolate the effects of text highlights from those of the interactive text-graph linking. All data-relevant textual segments referencing trends shown in the graphs were highlighted in Groups 2 and 3. A demographic analysis showed a balanced representation of gender across groups.

Participants began with a pre-questionnaire on Qualtrics, where they first provided their consent. They were then redirected to the version of the online article that we recreated and hosted on GitHub. After exploring the article, participants could click a link on the article's page to return to Qualtrics and complete a post-questionnaire, which included questions about usability and evaluations of their reading experience and loyalty intentions. We measured usability of the reading experience using the AttrakDiff questionnaire [19], which captures four dimensions of user experience: 1) Attractiveness (ATT), which reflects the overall appeal of the interface, 2) Pragmatic Quality (PG), which measures perceived usability and task efficiency, 3) Hedonic Quality – Stimulation (HQ-S), assessing how exciting and stimulating the experience is, and 4) Hedonic Quality – Identity (HQ-I), capturing how well the product aligns with the user's self-image. Loyalty was assessed using elements inspired by the E-S-QUAL scale [30]. This included three questions evaluating the user's intention to return to the website, recommend it to a friend, or pay for access. All measures were rated on a 7-point Likert scale, ranging from "Strongly disagree" to "Strongly agree", except the willingness to pay, which was measured using a 4-item scale ranging from "No" to "Yes" to reflect the categorical nature of the pricing preference question. Additionally, we collected user comments about their experience using an open-ended question.

5 RESULTS

To verify our first hypothesis (H1), we focus on the reader experience results captured by the AttrakDiff questionnaire. As shown in Table 1, we conducted separate one-way ANOVAs to compare each dimension across the three groups (Control, Highlights, Vizualink). The results revealed no significant group differences for Attractiveness (ATT), Hedonic Quality – Identity (HQ-I), or Pragmatic Quality (PG), suggesting similar perceptions of overall appeal, personal relevance, and usability across conditions. In contrast, significant differences emerged in Hedonic Quality - Stimulation (HQ-S), where the Vizualink group rated the experience as more stimulating compared to the Control and Highlights groups ($p < .001$).

To further examine the group differences observed in Hedonic Quality - Stimulation (HQ-S), especially in terms of effect sizes, a Tukey HSD test was performed. The results are presented in Table 2 and visualized in Figure 2. The test revealed that the difference in HQ-S scores between the Control group and the Highlights group

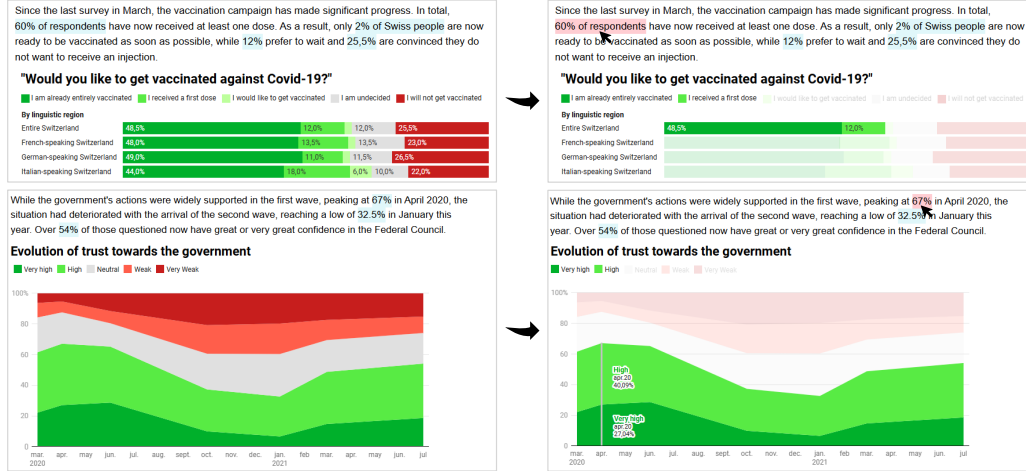


Figure 1: Integration of Vizualink in a data article. Screenshots on the left side display article sections with highlighted data points. When a reader hovers over a data point, the corresponding section of the relevant graph is emphasized, as shown in screenshots on the right side.

Table 1: Group Comparisons of AttrakDiff and Loyalty Intentions (One-Way ANOVA Results). * $p < .05$, ** $p < .01$, *** $p < .001$.

Dimension	Group Means (SD)						Statistical Test	
	Control		Highlights		Vizualink		F-value	p-value
Attractiveness (ATT)	5.48	(0.96)	5.37	(1.14)	5.48	(1.09)	0.39	0.676
Hedonic Quality – Stimulation (HQ-S)	4.81	(1.19)	4.66	(1.30)	5.33	(1.25)	9.31	0.000***
Hedonic Quality – Identity (HQ-I)	5.20	(0.92)	5.04	(0.98)	5.29	(1.09)	1.93	0.146
Pragmatic Quality (PG)	5.37	(0.98)	5.33	(1.04)	5.37	(1.08)	0.05	0.949
Willingness to pay	2.02	(0.92)	2.26	(0.96)	2.23	(1.02)	2.28	0.104
Intention to recommend	4.97	(1.51)	4.94	(1.66)	5.25	(1.47)	1.41	0.246
Intention to come back	4.83	(1.44)	4.89	(1.60)	5.19	(1.45)	1.94	0.146

was small and not statistically significant ($d = 0.122, p = 0.61$). However, a moderate effect size was observed between the Highlights group and the Vizualink group ($d = -0.524$), with the Highlights group scoring lower, and this difference being statistically significant ($p < 0.001$). Additionally, the Control group and the Vizualink group also showed a moderate effect size ($d = -0.421$), with the Vizualink group scoring higher, and this difference being statistically significant ($p = 0.004$). H1 is partially supported, as Vizualink significantly enhanced one of the three reading experience dimensions.

Table 2: Post-Hoc Comparisons of HQ-S Scores Between Groups (Tukey HSD with Cohen's d)

Group 1	Group 2	meandiff	p-adj	cohen _d
Control	Highlights	-0.15	0.61	0.12
Control	Vizualink	0.51	0.004	-0.42
Highlights	Vizualink	0.67	0.0001	-0.52

To evaluate how the improved reader experience was linked to loyalty intentions (willingness to pay, intention to recommend, and intention to come back)—H2—we constructed a structural model to study the impact of the treatment condition (Vizualink) on usability and the impact of the latter on these behavioral intentions. As shown in Table 1, the behavioral intentions did not differ significantly across groups. However, our focus was not on direct group comparisons but on understanding how Vizualink influences the usability of the reading experience, which in turn shapes behavioral

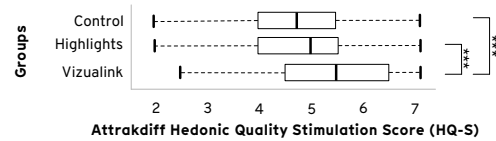


Figure 2: Boxplots of HQ-S scores. Scores range from 1 to 7, but all observed values fell between 2 and 7. *** $p < .001$.

intentions. To validate the structural model, we used the partial least squares (PLS) analysis technique with SmartPLS (version 4.1.0.6), a well-regarded variance-based approach frequently applied in various fields (e.g., information systems [11], HCI [28, 29, 6], marketing [34, 35]). PLS operates on a path model, represented in a diagram that illustrates the hypotheses and relationships between variables, which are then estimated using structural equation modeling (SEM) analysis [18]. The treatment condition was coded as a dummy variable with 0 for the Control and the Highlights conditions and 1 for the Vizualink condition. For hypothesis testing, we utilized T-statistics to assess the standardized path coefficients (β). The PLS analysis incorporated bootstrapping with 5000 resamples to ensure the robustness of the results. Figure 3 presents all significant links of the full model.

The PLS results are in line with the results presented above and show that Vizualink significantly improves HQ-S, but does not significantly affect the other dimensions. In addition, the results show that HQ-S is significantly and positively linked to all behavioral intentions, and PG is positively linked to the willingness to pay and

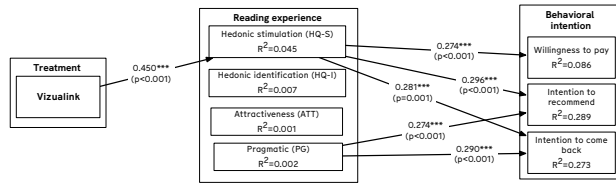


Figure 3: PLS results

the *intention to come back*. The other usability dimensions are not significantly linked to any behavioral intention. H2 is also partially supported as two out of the four reading experience dimensions are positively associated with most or all loyalty measures.

6 THEMATIC ANALYSIS OF PARTICIPANT COMMENTS

We performed a thematic analysis of participant comments from the Vizualink group using [8]’s approach. Through collaborative coding and iterative discussions, we identified four primary themes that reflect both the strengths and areas for improvement of Vizualink. These themes are elaborated below. The quotes included below were translated from French into English using DeepL [13]. The translations were reviewed for accuracy by the authors to ensure they reflect the original meaning.

Novelty and Unfamiliarity. Users frequently highlighted the novelty of Vizualink, appreciating its innovative approach to linking textual and graphical elements. Many participants expressed positive impressions of this new interaction type, emphasizing its role in facilitating understanding and memorization. For example, one user remarked, “*My overall impression is positive, I liked the idea, I’d never seen it before.*” Another participant noted, “*Actually, it’s not very common, but I really appreciate it. It makes the task easier and offers a lot more information.*”

Enhanced User Interaction. Participants largely appreciated the way Vizualink enhanced their interaction with data-driven articles. They highlighted its potential to make information more accessible, aid comprehension, and support recall. One user remarked, “*It’s nice to be able to link text and graphics visually without any extra effort.*” Participants also noted that the interaction encouraged deeper reflection, as another user reflected, “*Very interesting, it creates a back-and-forth between the reading and the graphic that provokes thought beyond what is written.*” Additionally, the feature was found to be suitable for both experts and novices, as one participant explained, “*Being an economist, it was easy for me to interpret the text and graph without help, but this system made reading more fluid[...] Moreover, I showed the article to my mother (a layperson in statistics) and she was more or less able to grasp the main idea.*”

Challenges and Criticisms. Despite its benefits, some participants raised criticisms about the feature. A recurring issue was that the feature’s utility diminished with simpler graphics. As one user observed, “*This is more pleasant, but doesn’t do much for simple graphs.*” Others found the interaction to be distracting or unnecessary, with comments such as “*The links linked to the graph were a little distracting when reading the article.*” Some participants also expressed concern that the feature might hinder active engagement, as one noted, “*Perhaps this removes the need for the user to be active in the search for information... perhaps it would therefore limit the memorization of information?*”

Suggestions for Improvement. Participants provided constructive feedback for improving Vizualink. Several users criticized the use of gray text and suggested alternative design choices. For example, one user stated, “*I think it’s an interesting idea, but you can do better in terms of execution. Putting all the text in gray*

is very ugly.” Another suggested improving the visibility of highlighted sections in the graphs, stating, “*Often the section of the graphic that is highlighted when you hover over the text is not visible on the page, so it’s confusing.*” Screen size limitations were also noted as a challenge, with one participant explaining, “*When I’m on a small screen, it was difficult if not impossible to highlight the text and get a full view of the graphic.*”

7 DISCUSSION AND CONCLUSION

The findings of this study suggest that incorporating an interactive text-graph integration feature, such as Vizualink, can enhance specific aspects of the reading experience of data stories, particularly hedonic stimulation (H1). Specifically, the AttrakDiff results revealed that the Vizualink group reported higher Hedonic Quality – Stimulation (HQ-S) compared to the Control and Highlights groups. Other usability dimensions, such as attractiveness, identity, and pragmatic quality, showed no significant group differences. These findings align with prior research emphasizing the role of interactivity in fostering the hedonic dimension of the reading experience [10, 1]. Thematic analysis confirmed these results, revealing that Vizualink offered an engaging way to interact with data-driven articles, with participants emphasizing its ability to seamlessly link text and graphs. Participants’ comments further highlighted that they appreciated how the feature encouraged deeper reflection and catered to both expert and novice readers, fostering a more inclusive reading experience, a key driver of reader loyalty [16]. However, it is also possible that part of this positive response stems from a novelty effect, as the feature was unfamiliar to participants; future studies should explore whether the perceived benefits persist with repeated exposure. Overall, these findings still provide evidence for the relevance of implementing and evaluating text-graph linking features in data-driven articles within real-world production settings, such as online newspapers.

Furthermore, whereas this study looked at participants in general, future studies could more specifically aim at understanding how different groups of readers navigate data stories and how Vizualink caters to each of them. For instance, it would be useful to understand if readers with lower data literacy can benefit from this feature to overcome potential issues with understanding graphs. Such results could further highlight how much scaffolding readers need or enjoy when reading data stories [25]. Feedback from participants can be used to fine-tune the design, including improving text visibility, ensuring highlighted graph sections remain visible on-screen, and addressing limitations on smaller devices. Future work should explore whether Vizualink improves data understanding or memorability, based on these improvement suggestions.

Moreover, the PLS analysis showed that the reading experience metrics significantly influenced loyalty measures (H2), including participants’ intentions to return, recommend, and pay for similar content, in line with prior work [23, 14]. This also serves as an encouraging indicator that such interactive features could strengthen user loyalty in real-world applications, such as online newspapers.

While this study used the AttrakDiff questionnaire and loyalty metrics, which provide valuable insights, these measures may not fully encompass all dimensions of reading experience and loyalty, and might be subject to self-report biases. Additionally, the study does not consider how a focus on usability and loyalty might trade off with other goals, such as cognitive or affective learning, which are also central to communicative visualizations. Future research could address these limitations by expanding to more diverse samples and testing on more data-driven articles, as well as exploring additional metrics. Finally, a future study would benefit from an ecologically valid setting with online news readers by implementing the feature on pilot articles with a news media organization.

SUPPLEMENTAL MATERIALS

Supplemental materials are included with this submission. They comprise: (1) the implementation of the Vizualink feature, and (2) the dataset used for analysis.

ACKNOWLEDGEMENTS

This research was partially funded by the Swiss National Science Foundation (SNSF) under the NRP 80 project 408040.210123.

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